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Dip. In Engg. (4th Semester)

Examination, May-June, 2013

Branch : Mech.

MATERIAL TECHNOLOGY

Time Allowed : Three Hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : (i) All questions are compulsory unless mentioned otherwise.

(ii) In case of any doubt or dispute, the English version question should be treated as final.

Q. 1. Explain any five mechanical properties : 10

(a) Ductility

(b) Brittleness

(c) Malleability

(d) Toughness

(e) Hardness

(f) Weddability

(g) Strength

Or

Explain destructive and non-destructive testing of material.

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Q. 2. (a) Explain BCC, FCC and HCP Crystal Structure. 5

(b) Explain with neat sketch of crystal structural imperfections. 5

Q. 3. (a) Explain Recovery, Recrystallization and Grain Growth. 5

(b) Explain metallurgical microscope, use and care of microscope with diagram. 5

Q. 4. (a) Explain solid solutions. Write factors affecting the formation of solid solutions. 5

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(b) Explain iron-carbon equillibrium with neat sketch. 5

Q. 5. (a) Explain Annealing, Hardening, Normalizing and Tempering. 5

(b) Explain with neat sketch TTT curves and its uses. 5

Q. 6. (a) Define alloy steel. List any five common alloying elements of steel. 5

(b) Name of different types of cast iron. How does presence of graphite influences its properties. 5

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Q. 7. (a) Write the important properties of copper, which make it an important engineering material. 5

(b) List the properties of aluminium 5

Q. 8. (a) What are the refractories and explain its properties ? 5

(b) Explain vulcanization process of rubber. 5

Q. 9. (a) List the advantages and application of compacting and sintering. 5

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(b) What is powder metallurgy ? List atleast five advantages of powder metallurgy. 5

Q. 10. What is corrosion ? List and explain any three method of minimizing corrosion. 10

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